

THE COEFFICIENT OF DETERMINATION

The coefficient of determination is another measure that helps us to get some information about the correlation between two variables. The coefficient of determination is used to inform us how well our regression line actually represents the data set under consideration.

Consider the following data set comparing the variables student height and student weight.

Height (cm)	160	172	173	180	166	187	172	191	158	164	186	156
Weight (kg)	52.3	54.8	53.4	58.2	56.8	62.6	53.4	62.8	45.8	50.1	69.2	49.4

Using a calculator the following statistics may be displayed:

For the given data set the correlation coefficient, r , is 0.8874051.

For the given data set the coefficient of determination, r^2 , is 0.7874874.

The coefficient of determination is simply the square of the correlation coefficient. That is $r^2 = (0.8874051)^2 = 0.7875$ rounded to 4 d.p.

Linear Reg

$$y = ax + b$$

$$a = 0.4948204$$

$$b = -29.41701$$

$$r = 0.8874051$$

$$r^2 = 0.7874878$$

The coefficient of determination gives us the variation in the response or dependent variable that is directly related to or explained by the variation in the explanatory or the independent variable.

For our example the coefficient of determination gives the proportion of the variation in weight that is directly related to the variation in the height of these students.

If we change the value of r^2 into a percentage, that is $r^2 = 0.7875 \times 100\% = 78.75\%$, then this tells us that approximately 79% of the variation in the weight of these students can be accounted for or explained by the variation of the height of these students. This indicates that there is a strong association or link between the weight and height of these students.

Now this infers that the percentage of unexplained variation is given by $100\% - 78.75\% = 21.25\%$. This tells us that approximated 21% of the variation in weight is unexplained by the variation in height.

NOTE: $1 - r^2$ is known as the **coefficient of non-determination**. The coefficient of non-determination measures the percentage of the variation in the response variable which is explained by **chance** and **other factors** or the percentage of variation in the response variable NOT explained by variation in the explanatory variable.

Correlation Coefficient and Coefficient of Determination

The **correlation coefficient**, r or r_{xy} , is also called Pearson's Correlation Coefficient.

The correlation coefficient is a measure of linear association between two variables and it gives us a measure of the **strength** and **direction** of a linear relationship.

The value of the correlation coefficient for a data set always lies in the interval $-1 \leq r \leq 1$.

The **coefficient of determination**, r^2 , the square of the correlation coefficient, is a measure of how much of the variation in the **response** or **dependent variable** can be explained by the variation in the **explanatory** or **independent variable**.

The value of the coefficient of determination for a data set always lies in the interval $0 \leq r^2 \leq 1$, because squaring any negative number results in a positive result.

For a correlation coefficient of $r = 0.5$ the coefficient of determination is $r^2 = (0.5)^2 = 0.25$. Hence a correlation of 0.5 implies that 25% of the variation in the response variable is accounted for, or explained by the variation in the explanatory variable.

The coefficient of determination is a more accurate measure of the strength of relationship than the correlation coefficient, especially if we are making comparisons.

For a data set with $r = 0.8$ we would think that the strength of this relationship would be twice as strong as that of a data set with $r = 0.4$. However this is not so because an r value of 0.8 indicates a relationship that is four times as strong as one with an r value of 0.4. This is because examination of r^2 for an r value of 0.8 is 0.64 and the r^2 for an r value of 0.4 is only 0.16. Stating the obvious, 0.64 is four times 0.16.

By definition, the coefficient of determination, r^2 , can be found from the correlation coefficient by squaring the correlation coefficient. For example a data set with $r = 0.7$ has a coefficient of determination (r^2) of $(0.7)^2 = 0.49$ or 49%.

The correlation coefficient, r , can be determined given the coefficient of determination.

For example if $r^2 = 0.81$, then $r = \pm\sqrt{0.81} = \pm 0.9$. Now to determine whether the r value is positive or negative we need to examine either the scatterplot of the data set or the equation of the regression line.

The scatterplot will identify whether the direction of the data set is positive or negative.

Identifying the gradient of the regression line will tell us whether the correlation coefficient, r , is positive or negative. If the gradient of the regression line is positive so also is the value of r , and if the gradient of the regression line is negative then the value of r is also negative.

Properties of the Coefficient of Determination

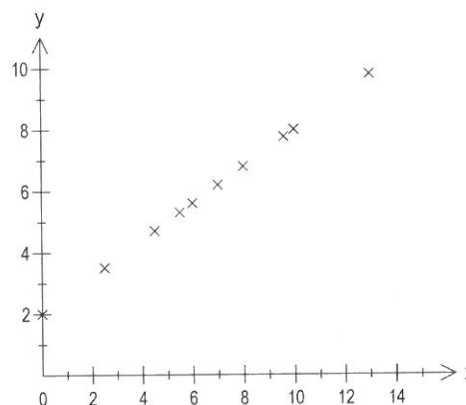
- because the coefficient of determination, r^2 , is the result of squaring the correlation coefficient, r , the coefficient of determination can never be negative, thus $0 \leq r^2 \leq 1$,
- $r^2 = 1$ indicates that the regression line perfectly fits the data, that is all the data points line on the regression line.
When $r^2 = 1$ then this means that 100% of the variation in the response or dependent variable can be explained by the variation in the explanatory or independent variable.
- $r^2 = 0$ indicates that the regression line does not fit the data at all. This may be because the data is not linear, or the data is random. In such situations the regression line is a horizontal line, (i.e. slope is zero) and the y intercept is the mean of the values of the response or dependent variable.
When $r^2 = 0$ then this means that none or 0% of the variation in the response or dependent variable can be explained by the variation in the explanatory or independent variable.
- $0 < r^2 < 1$ indicates that the regression line fits the data to some degree, the closer r^2 is to 1 the better the goodness of fit or in other words the more the variation in the response or dependent variable can be explained by the variation in the explanatory or independent variable.
As r^2 gets closer to 1 the values of the response or dependent variable get closer to the regression line, and as r^2 gets closer to 0 the values of the response or dependent variable get further from the regression line.
If $r^2 = 0.8$, then this means that 80% of the variation in the response or dependent variable can be explained by the variation in the explanatory or independent variable.
The remaining 20% of the total variation in the response or dependent variable remains unexplained.
- A high value of r^2 does not imply causality. When we examine the value of r^2 we may conclude that the two variables are related but the value of r^2 does not tell us that changes in one variable cause the changes in the other.

Example 1

Consider the following bivariate data set.

x	6	10	7	13	4.5	8	9.6	2.5	0	5.5
y	5.6	8	6.2	9.8	4.7	6.8	7.76	3.5	2	5.3

- Construct a scatter plot and comment on the relationship between x and y .
- Determine the value of the correlation coefficient.
- Describe the data using your value of r .
- Determine the value of the coefficient of determination.
- Interpret the coefficient of determination.
- Determine the equation of the least squares regression line that models the relationship between x and y .
- Identify and interpret the slope of the least squares regression line in this context.



Solution

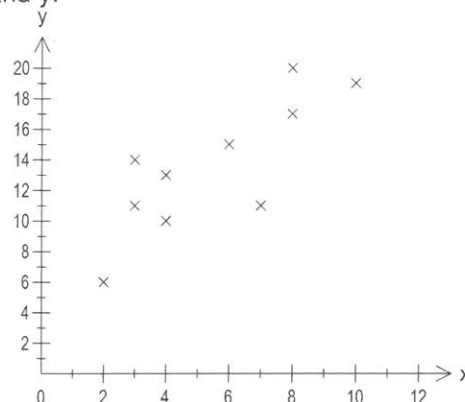
- The scatter graph shows a perfect positive linear relationship between x and y .
- Using a calculator, $r = 1$
- There is a perfect positive linear relationship between x and y .
- Now $r = 1$ hence $r^2 = (1)^2 = 1$. Coefficient of determination is 1.
- 100% of the variation in y can be explained by the variation in x .
- The equation of the regression line is $\hat{y} = 0.6x + 2$
- The slope is 0.6. The slope of the regression line predicts that on average we can expect an increase of 0.6 units in y for a one unit increase in x .

Example 2

Consider the following bivariate data set.

x	8	4	3	6	7	3	2	4	8	10
y	17	13	14	15	11	11	6	10	20	19

- Construct a scatter plot to show the relationship between x and y.
- Describe the relationship shown in the scatter plot.
- Determine the value of the correlation coefficient. Round your answer to 4 decimal places.
- Describe the data using your value of r.
- Determine the value of the coefficient of determination.
- Interpret the coefficient of determination.
- Determine the equation of the least squares regression line that models the relationship between x and y. Give the regression line parameters rounded to two decimal places.
- Identify and interpret the gradient of the least squares regression line in this context.

**Solution**

- See graph.
- The relationship between x and y can be described as a moderate to strong positive linear relationship.
- Using a calculator, $r = 0.8060$ (4 d.p.)
- There is a strong positive linear relationship between x and y.
- Now $r = 0.8060$ hence $r^2 = (0.8060)^2 = 0.6496$ (4 d.p.)
Coefficient of determination is 0.6496
- 64.96% of the variation in y can be explained by the variation in x.
- The equation of the regression line is $\hat{y} = 1.30x + 6.44$.
- The gradient is 1.30. The gradient of the regression line predicts that on average we can expect an increase of 1.30 units in y for a unit increase in x.

Example 3

Data was collected by a maths teacher to investigate the relationship between student test marks and the number of hours spent studying for the test. The collected data is displayed in the table below where the test marks have been recorded as percentages.

Number of hours of study	8	4	2	6	6	4	5	5	7	8	3	9
Test mark (%)	88	62	28	62	70	55	62	70	80	79	44	96

- Identify the explanatory and response variables.
- View, on your calculator or computer, the scatter plot of this data and describe the relationship between the variables.
- Determine the value of the coefficient of determination. Give your answer rounded to 3 decimal places.
- Interpret the coefficient of determination.
- Give the equation of the regression line that models this data. Round parameters to two decimal places.
- Is the linear regression model found in (e) appropriate for this data set? Justify your response.

Solution

- Explanatory variable is the number of hours of study and the response variable is the test mark given as a percentage.
- The scatter plot shows a linear relationship between these variables which is positive and strong.
- From the calculator $r^2 = 0.909$.
- 90.9% of the variation in the test mark percentage can be explained by the variation in the number of hours of study.
- From the calculator $y = 8.32x + 19.88$. Using the correct variable names that model this data the required equation is given by: Test mark = $8.32 \times$ number of hours of study + 19.88
- Yes the model is appropriate. The regression model is appropriate because the relationship between the variables is linear and strong with a high value of r^2 . The value of r^2 informs us that only 9.1% (100 – 90.9) of the variation between the variables is unexplained by the model.

Example 4

The correlation coefficient for a bivariate data set was found to be -0.75. Find and interpret the coefficient of determination.

Solution

$r^2 = (-0.75)^2 = 0.5625$. Coefficient of determination is 0.5625.

This coefficient of determination tells us that 56.25% of the variation in the response variable can be explained by the variation in the explanatory variable.

Example 5

The least squares regression line $y = 11.4 - 0.35x$ was calculated using latitude measured in degrees and average January temperature in degrees Celsius for 25 cities. The value of the coefficient of determination was given as 0.7257 correct to 4 decimal places.

- Identify the explanatory and response variables.
- What does the coefficient of determination tell you about these variables.
- Calculate the value of Pearson's correlation coefficient.

Solution

- Latitude (x) is the explanatory variable average January temperature (y) is the response variable.
- 72.57% of the variation in average January temperature is explained by the variation in latitude.
- $r^2 = 0.7257$, hence $r = \pm\sqrt{0.7257} = \pm 0.8519$ (4 d.p.)

Now the gradient of the regression line is negative hence the r value is also negative.

Therefore Pearson's correlation coefficient is -0.8519

EXERCISE 3B

- For each of the following:
 - Calculate the coefficient of determination for each of the following values of r giving your answer correct to 3 decimal places.
 - Interpret the coefficient of determination.

(a) $r = 0.58$

(b) $r = -0.709$

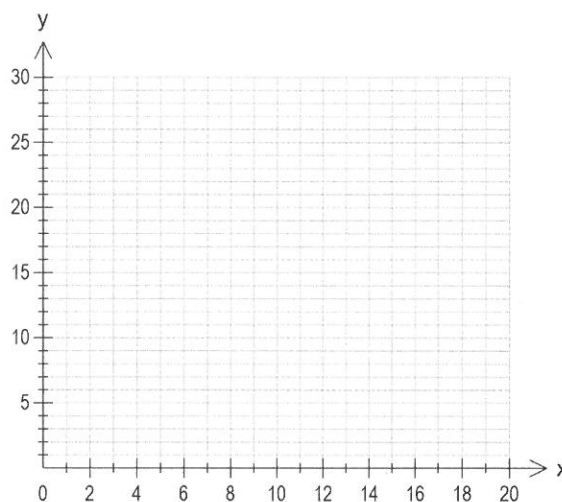
(c) $r = 0.421$

(d) $r = -0.8999$

- Consider the following bivariate data set.

x	6	13	10	8	18	4	17	5	2	15	17	9
y	16	20	25	14	28	9	29	8	11	27	24	20

- Construct a scatter plot and comment on the relationship between x and y .



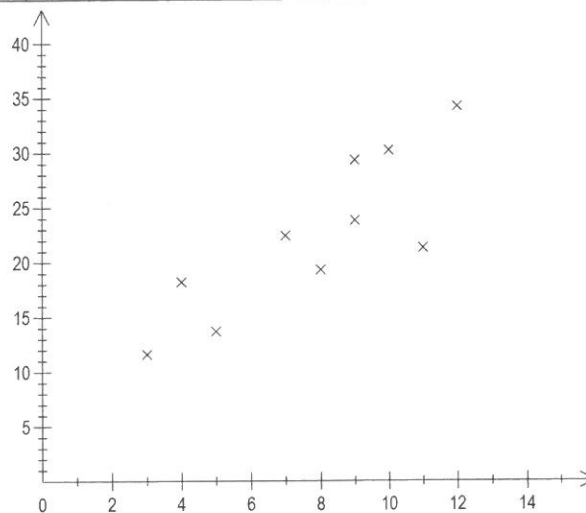
- Determine the value of the correlation coefficient.
- Describe the data using your value of r .
- Determine the value of the coefficient of determination.
- Interpret the coefficient of determination.

- Determine the equation of the least squares regression line that models the relationship between x and y .
- Identify and interpret the slope of the least squares regression line.
- Is the linear regression model appropriate? Justify your answer.

3. The following table gives the age of cars, in years and the cost of repairs in hundreds of dollars of 15 cars of a particular model.

Age of car	12	9	11	8	10	4	3	9
Cost of repairs (\$'00)	34.2	23.8	21.3	19.3	30.2	18.2	11.6	29.3
Age of car	7	5	14	11	6	10	12	
Cost of repairs (\$'00)	22.4	13.7	30.4	29.9	19.0	28.0	27.1	

- (a) Identify the explanatory and response variables.
- (b) Complete the scatter plot of the given data by labelling the axes and adding the five points tabled in **bold**.
- (c) Determine, correct to 4 decimal places the coefficient of determination.
- (d) Interpret the coefficient of determination.

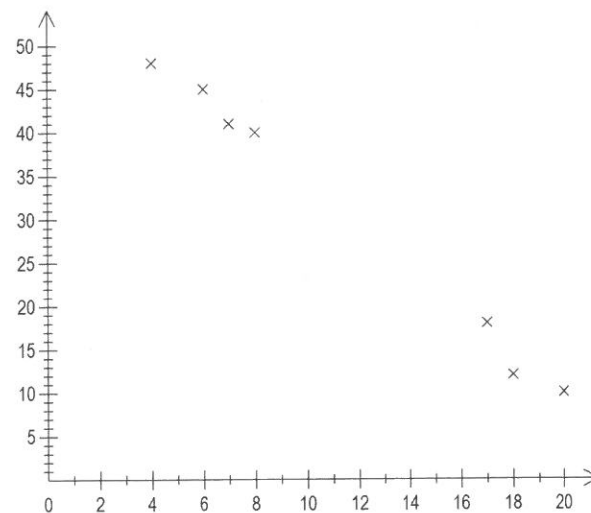


- (e) Find the equation of the least squares regression line that models this data and add it to the scatter plot.
- (f) Is the model found in (e) appropriate? Justify your response.

4. The table shows the number of widgets purchased at one time and the cost per widget.

Number purchased	4	6	7	8	9	10	12	14	15	17	18	20
Cost per widget (cents)	48	45	41	40	38	34	32	22	20	18	12	10

- (a) Identify the explanatory and response variables.
- (b) Complete the scatter plot of the given data by labelling the axes and adding the five points tabled in **bold**.
- (c) Determine, correct to 4 decimal places the coefficient of determination.
- (d) Interpret the coefficient of determination.



- (e) Find the equation of the least squares regression line that models this data and add it to the scatter plot.
- (f) Identify and interpret the slope of the least squares regression line.
- (g) Identify and interpret the y intercept.
- (h) Is the model found in (e) appropriate? Justify your response.

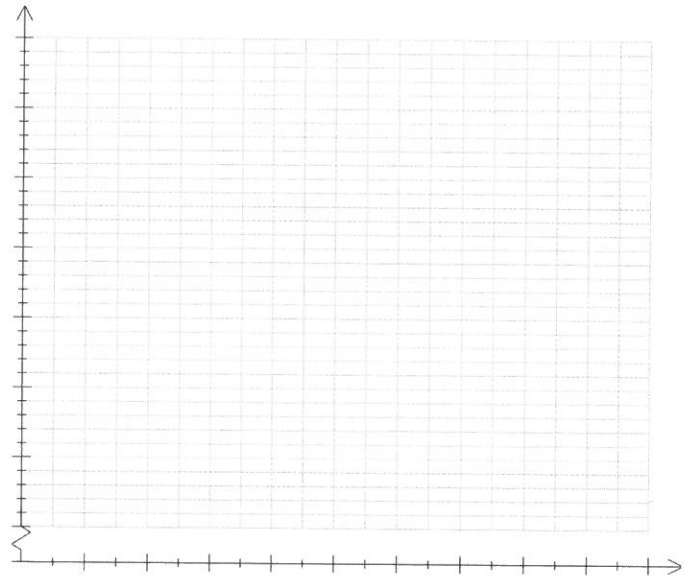
5. The number of hours students spent playing video games and their History test marks (%) were recorded for a group of Year 12 students. The recorded results are displayed in the table below.

History(%)	72	53	94	36	91	90	68	43	86	72	69	85
Video game hours	6	9	1	17	3	0	5	18	7	5	10	5

- (a) Identify the explanatory and response variables.

- (b) On the given axes construct a scatter plot to represent these variables.
 (c) Describe the relationship between these variables shown in the scatter plot.

- (d) State the value of Pearson's correlation coefficient and interpret it.



- (e) State the coefficient of determination and interpret it.
 (f) Find the equation of the regression line that models the data.
 (g) Identify and interpret the gradient of the least squares regression line.
 (h) Identify and interpret the y-intercept of the least squares regression line.
 (i) Add the regression line to your scatter plot.
 (j) Is this model appropriate for the given data set? Justify your answer.

6. Mike is conducting research on monthly expenses for medical care. He collected data from 578 families noting the monthly expense, in dollars, and the number of people in each of these families. On entering the data into his calculator the following screen was displayed

- (a) Identify the explanatory and response variables.
 (b) Write the equation of the least squares regression line in the context of Mike's research.
 (c) What percentage of the variation in medical expenses is explained by the size of the family?
 (d) Is the linear model that Mike is considering based on the calculator display appropriate? Justify.

Linear Reg

$$y = ax + b$$

a = 16.831330
 b = 110.472056
 r = 0.695421
 $r^2 = 0.483610$

7. The regression model for a data set is given by $\hat{y} = 467.5 - 12.8x$. If the coefficient of determination for the data set is 0.687, what is the value of the correlation coefficient? Give your answer rounded to four decimal places.
8. The value of the correlation coefficient is reported by a researcher to be $r_{ab} = -0.5$. Determine the coefficient of determination for this data and explain what it means.

9. Medical research data on length of the left forearm (cm) and height (cm) of a group of students is displayed in the table below.

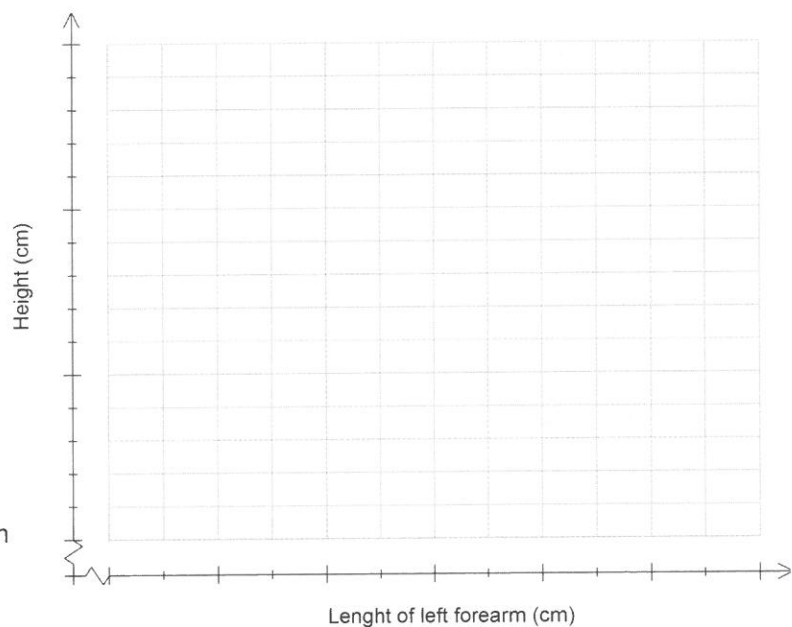
Left forearm length (cm)	22.8	29.1	24.3	20.4	21.8	27.2	25.4	28.4	30.6	22.1	20.6	30.7
Height (cm)	169	178	173	167	174	176	176	180	184	169	171	182

- (a) Identify the response and explanatory variables.

- (b) Construct a scatter plot to represent the data.

- (c) Determine the value of the correlation coefficient and interpret it.

- (d) Determine the value of the coefficient of determination and interpret it.



- (e) Find the equation of the least squares regression line that models the data.

- (f) Find the gradient of the regression line and explain the gradient in context of the variables.

- (g) Find the y-intercept of the regression line and explain the y-intercept in context of the variables.

- (h) Add the regression line to your scatter plot.

- (i) Is the linear model appropriate for this data. Justify your response.

10. If the coefficient of correlation is 0.90, what is the percentage of the variation in the response variable explained by the variation in the explanatory variable?
11. The least squares regression line is given by $\hat{y} = 1.4x - 55.3$. If the coefficient of determination is 0.49, what is the coefficient of correlation?